# V4.7 BOM 核对清单（按位号顺序）

共 229 个位号，其中 207 个需要贴片。 贴完一个在左侧方框打勾。坐标以 PCB 左上角为原点，单位 mm。

有极性/方向 —— 贴反不工作　不贴

| 贴 | 位号 | 值 / 型号 | 封装 | 位置 (x, y) | 阶段 | 备注 |
|---|---|---|---|---|---|---|
| | C1 | 10uF | C_0805_2012Metric | 70.8, 96.5 | A 电源 | |
| | C2 | 1uF | C_0603_1608Metric | 85.9, 99.1 | A 电源 | |
| | C3 | 1uF | C_0603_1608Metric | 70.8, 98.9 | A 电源 | |
| | C4 | 1uF | C_0603_1608Metric | 78.7, 97.8 | A 电源 | |
| | C5 | 1uF | C_0402_1005Metric | 85.9, 96.7 | A 电源 | |
| | C6 | 1uF | C_0603_1608Metric | 92.1, 97.9 | A 电源 | |
| | C10 | 100nF | C_0603_1608Metric | 61.0, 92.6 | C 传感器 + GNSS | |
| | C11 | 100nF | C_0603_1608Metric | 64.6, 87.2 | C 传感器 + GNSS | |
| | C12 | 100nF | C_0603_1608Metric | 56.5, 87.7 | C 传感器 + GNSS | |
| | C13 | 100nF | C_0603_1608Metric | 59.1, 92.6 | C 传感器 + GNSS | |
| | C14 | 100nF | C_0603_1608Metric | 64.6, 89.9 | C 传感器 + GNSS | |
| | C15 | 4.7uF | C_0603_1608Metric | 59.9, 99.3 | C 传感器 + GNSS | |
| | C16 | 100nF | C_0603_1608Metric | 64.6, 92.5 | C 传感器 + GNSS | |
| | C17 | 100nF | C_0603_1608Metric | 71.4, 81.0 | C 传感器 + GNSS | |
| | C18 | 10uF | C_0805_2012Metric | 74.4, 80.8 | C 传感器 + GNSS | |
| | C19 | 15pF C0G | C_0603_1608Metric | 88.4, 69.1 | B MCU + Flash | |
| | C20 | 15pF C0G | C_0603_1608Metric | 79.8, 69.1 | B MCU + Flash | |
| | C21 | 100pF | C_0402_1005Metric | 103.6, 59.7 | E 1090 接收链 | |
| | C22 | 100nF | C_0603_1608Metric | 93.2, 75.8 | B MCU + Flash | |
| | C23 | 100nF | C_0603_1608Metric | 91.8, 81.5 | B MCU + Flash | |
| | C24 | 100nF | C_0603_1608Metric | 80.9, 80.2 | B MCU + Flash | |
| | C25 | 100nF | C_0603_1608Metric | 83.2, 71.6 | B MCU + Flash | |
| | C26 | 100nF | C_0603_1608Metric | 78.9, 74.6 | B MCU + Flash | |
| | C27 | 100nF | C_0603_1608Metric | 77.9, 82.5 | B MCU + Flash | |
| | C28 | 1uF | C_0603_1608Metric | 83.1, 83.3 | B MCU + Flash | |
| | C29 | 1uF | C_0603_1608Metric | 80.9, 83.3 | B MCU + Flash | |
| | C30 | 100pF | C_0402_1005Metric | 102.8, 67.5 | E 1090 接收链 | |
| | C31 | 100pF C0G | C_0402_1005Metric | 111.3, 62.2 | E 1090 接收链 | |
| | C32 | 100pF C0G | C_0402_1005Metric | 114.3, 62.2 | E 1090 接收链 | |
| | C33 | 100pF C0G | C_0402_1005Metric | 120.4, 63.2 | E 1090 接收链 | |
| | C34 | 100pF C0G | C_0402_1005Metric | 117.1, 73.4 | E 1090 接收链 | |
| | C35 | 100pF C0G | C_0402_1005Metric | 115.1, 71.1 | E 1090 接收链 | |
| | C36 | 100pF | C_0402_1005Metric | 106.0, 64.3 | E 1090 接收链 | |
| | C37 | 100nF | C_0402_1005Metric | 113.9, 77.0 | E 1090 接收链 | |
| | C38 | 100pF | C_0402_1005Metric | 111.6, 77.0 | E 1090 接收链 | |
| | C39 | 100pF | C_0402_1005Metric | 118.2, 92.8 | D 978 收发 | |
| | C40 | 3.6pF | C_0402_1005Metric | 110.7, 94.2 | D 978 收发 | |
| | C41 | 2.7pF | C_0402_1005Metric | 115.9, 97.5 | D 978 收发 | |
| | C42 | 6.2pF | C_0402_1005Metric | 115.9, 95.2 | D 978 收发 | |
| | C43 | 3pF | C_0402_1005Metric | 115.9, 92.8 | D 978 收发 | |
| | C44 | 3.6pF | C_0402_1005Metric | 112.8, 96.5 | D 978 收发 | |
| | C45 | 100pF | C_0402_1005Metric | 110.7, 90.2 | D 978 收发 | |
| | C46 | 3pF | C_0603_1608Metric | 104.9, 78.5 | E 1090 接收链 | |
| | C47 | 200pF | C_0603_1608Metric | 104.9, 80.4 | E 1090 接收链 | |
| | C48 | 1uF | C_0402_1005Metric | 107.4, 64.3 | E 1090 接收链 | |
| | C49 | 100nF | C_0603_1608Metric | 76.7, 75.4 | E 1090 接收链 | |
| | C51 | 1nF | C_0603_1608Metric | 77.0, 81.0 | E 1090 接收链 | |

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|---|---|---|---|---|---|---|
| | C52 | 100nF | C_0402_1005Metric | 99.6, 82.3 | E 1090 接收链 | |
| | C53 | 100pF | C_0402_1005Metric | 93.6, 62.9 | E 1090 接收链 | J7测IFA要求后级断路；U17等未贴时C53可保留 |
| | C54 | 100pF | C_0402_1005Metric | 103.9, 61.7 | E 1090 接收链 | |
| | C55 | 100pF | C_0603_1608Metric | 97.6, 66.5 | E 1090 接收链 | |
| | C56 | 100pF | C_0603_1608Metric | 95.0, 66.5 | E 1090 接收链 | |
| | C57 | 100pF | C_0402_1005Metric | 68.3, 81.0 | C 传感器 + GNSS | |
| | C58 | 100pF | C_0402_1005Metric | 57.7, 76.5 | C 传感器 + GNSS | |
| | C59 | 100pF | C_0402_1005Metric | 57.6, 69.8 | C 传感器 + GNSS | |
| | C60 | 22uF | C_0805_2012Metric | 99.3, 98.0 | D 978 收发 | |
| | C61 | 100nF | C_0603_1608Metric | 102.7, 99.1 | D 978 收发 | |
| | C62 | 100nF | C_0603_1608Metric | 105.4, 98.3 | D 978 收发 | |
| | C63 | 100nF | C_0603_1608Metric | 107.1, 84.6 | D 978 收发 | |
| | C64 | 100nF | C_0603_1608Metric | 100.2, 85.8 | D 978 收发 | |
| | C65 | 100nF | C_0603_1608Metric | 99.0, 94.7 | D 978 收发 | |
| | C66 | 22uF | C_0805_2012Metric | 102.9, 97.2 | D 978 收发 | |
| | C67 | 1uF | C_0603_1608Metric | 103.6, 85.8 | D 978 收发 | |
| | C68 | 18pF | C_0603_1608Metric | 113.2, 85.0 | D 978 收发 | |
| | C69 | 18pF | C_0603_1608Metric | 110.1, 85.0 | D 978 收发 | |
| | C70 | 47nF | C_0603_1608Metric | 135.7, 80.2 | A 电源 | |
| | C71 | 4.7uF(带电源版) | C_0603_1608Metric | 133.4, 79.0 | A 电源 | |
| | C72 | 10uF | C_1206_3216Metric | 130.9, 79.2 | A 电源 | |
| | C73 | 10uF | C_0805_2012Metric | 141.3, 88.1 | A 电源 | |
| | C74 | 10uF | C_0805_2012Metric | 143.3, 84.9 | A 电源 | |
| | C75 | 10uF | C_0805_2012Metric | 139.1, 88.1 | A 电源 | |
| | C76 | 22uF | C_1206_3216Metric | 135.2, 96.8 | A 电源 | |
| | C77 | 22uF 16V X5R | C_1206_3216Metric | 141.2, 99.7 | A 电源 | |
| | C78 | 1uF | C_0603_1608Metric | 134.3, 93.8 | A 电源 | |
| | C79 | 1uF | C_0603_1608Metric | 130.6, 65.0 | A 电源 | |
| | C80 | 1uF 25V | C_0603_1608Metric | 136.3, 74.6 | A 电源 | |
| | C81 | 470pF C0G | C_0402_1005Metric | 115.2, 64.5 | E 1090 接收链 | |
| | C82 | 100nF | C_0603_1608Metric | 91.4, 67.5 | B MCU + Flash | |
| | C83 | 100nF | C_0603_1608Metric | 91.4, 69.8 | B MCU + Flash | |
| | C84 | 1uF | C_0603_1608Metric | 83.3, 85.6 | B MCU + Flash | |
| | C85 | 100nF | C_0603_1608Metric | 113.7, 80.5 | B MCU + Flash | |
| | C86 | 100nF | C_0603_1608Metric | 121.4, 88.1 | B MCU + Flash | |
| | C87 | 10nF C0G | C_0603_1608Metric | 83.5, 88.5 | B MCU + Flash | |
| | C88 | 100pF | C_0603_1608Metric | 126.4, 90.7 | G 航空波段 AIRBAND | |
| | C89 | 18pF | C_0603_1608Metric | 128.7, 93.9 | G 航空波段 AIRBAND | |
| | C91 | 10uF | C_0805_2012Metric | 129.0, 108.2 | G 航空波段 AIRBAND | |
| | C92 | 10uF | C_0805_2012Metric | 121.3, 100.8 | G 航空波段 AIRBAND | |
| | C93 | 10uF | C_0805_2012Metric | 124.2, 109.4 | G 航空波段 AIRBAND | |
| | C94 | 100nF | C_0603_1608Metric | 120.5, 109.4 | G 航空波段 AIRBAND | |
| | C95 | 100nF | C_0603_1608Metric | 119.4, 100.1 | G 航空波段 AIRBAND | |
| | C96 | 100nF | C_0603_1608Metric | 128.0, 102.1 | G 航空波段 AIRBAND | |
| | C97 | 1uF | C_0603_1608Metric | 129.8, 102.1 | G 航空波段 AIRBAND | |
| | C98 | 100nF | C_0603_1608Metric | 131.6, 102.1 | G 航空波段 AIRBAND | |
| | C99 | 100nF | C_0402_1005Metric | 133.6, 100.1 | G 航空波段 AIRBAND | |
| | R1 | 10k | R_0603_1608Metric | 55.2, 84.5 | C 传感器 + GNSS | |
| | R2 | 4.7k | R_0603_1608Metric | 64.1, 97.8 | C 传感器 + GNSS | |

| 贴 | 位号 | 值 / 型号 | 封装 | 位置 (x, y) | 阶段 | 备注 |
|---|---|---|---|---|---|---|
| | **R3** | 4.7k | R_0603_1608Metric | 64.1, 95.2 | C 传感器 + GNSS | |
| | **R4** | 1k | R_0603_1608Metric | 82.0, 69.1 | B MCU + Flash | |
| | **R5** | 10k | R_0603_1608Metric | 80.0, 73.2 | B MCU + Flash | |
| | **R6** | 10k | R_0603_1608Metric | 87.9, 86.4 | B MCU + Flash | |
| | **R7** | 5.1k DNP(带电源版) | R_0603_1608Metric | 141.4, 62.5 | 不贴（DNP） | |
| | **R8** | 5.1k DNP(带电源版) | R_0603_1608Metric | 139.4, 62.5 | 不贴（DNP） | |
| | **R9** | 27R | R_0603_1608Metric | 87.9, 83.3 | B MCU + Flash | |
| | **R10** | 27R | R_0603_1608Metric | 85.3, 83.3 | B MCU + Flash | |
| | **R11** | 3.32k | R_0402_1005Metric | 105.1, 59.7 | E 1090 接收链 | |
| | **R17** | 10k | R_0603_1608Metric | 122.0, 97.4 | D 978 收发 | |
| | **R18** | 10k | R_0603_1608Metric | 91.4, 71.4 | E 1090 接收链 | |
| | **R19** | 52.3R | R_0402_1005Metric | 115.1, 73.4 | E 1090 接收链 | |
| | **R21** | 0R | R_0603_1608Metric | 105.0, 72.6 | E 1090 接收链 | |
| | **R22** | 0R | R_0603_1608Metric | 100.3, 66.5 | E 1090 接收链 | |
| | **R23** | 0R | R_0603_1608Metric | 93.2, 77.2 | E 1090 接收链 | |
| | **R24** | 0R DNP(旁路外接) | R_0402_1005Metric | 101.1, 63.1 | 不贴（DNP） | |
| | **R25** | 0R DNP(旁路板载) | R_0402_1005Metric | 95.5, 62.9 | 不贴（DNP） | |
| | **R26** | 10k | R_0603_1608Metric | 67.3, 73.7 | C 传感器 + GNSS | |
| | **R27** | 10k | R_0603_1608Metric | 67.3, 71.1 | C 传感器 + GNSS | |
| | **R30** | 1k DNP | R_0402_1005Metric | 96.8, 77.2 | 不贴（DNP） | |
| | **R31** | 1k | R_0603_1608Metric | 101.6, 75.7 | E 1090 接收链 | |
| | **R32** | 10k | R_0603_1608Metric | 104.9, 75.7 | E 1090 接收链 | |
| | **R33** | 100k | R_0603_1608Metric | 100.9, 82.7 | E 1090 接收链 | |
| | **R34** | 10k | R_0603_1608Metric | 77.0, 73.2 | E 1090 接收链 | |
| | **R35** | 10k | R_0603_1608Metric | 77.0, 78.4 | E 1090 接收链 | |
| | **R36** | 1k DNP | R_0603_1608Metric | 96.5, 81.8 | 不贴（DNP） | |
| | **R37** | 0R | R_0603_1608Metric | 137.0, 66.4 | A 电源 | |
| | **R38** | 180R | R_0603_1608Metric | 133.1, 88.2 | A 电源 | |
| | **R39** | 5.62k | R_0603_1608Metric | 136.2, 88.2 | A 电源 | |
| | **R40** | 31.6k | R_0603_1608Metric | 134.6, 88.2 | A 电源 | |
| | **R41** | 470k | R_0603_1608Metric | 139.1, 98.6 | A 电源 | |
| | **R42** | 150k | R_0603_1608Metric | 137.4, 98.6 | A 电源 | |
| | **R43** | 1k | R_0603_1608Metric | 127.5, 65.0 | A 电源 | |
| | **R44** | 10k | R_0603_1608Metric | 137.0, 69.5 | A 电源 | |
| | **R45** | 10k | R_0603_1608Metric | 129.2, 84.0 | A 电源 | |
| | **R46** | 10k | R_0603_1608Metric | 129.2, 86.5 | A 电源 | |
| | **R47** | 10k | R_0603_1608Metric | 98.5, 91.7 | D 978 收发 | |
| | **R48** | 0R | R_0603_1608Metric | 126.8, 67.0 | A 电源 | |
| | **R49** | 0R | R_0603_1608Metric | 126.8, 69.4 | A 电源 | |
| | **R50** | 30k | R_0603_1608Metric | 86.5, 92.8 | B MCU + Flash | |
| | **R51** | 10k | R_0603_1608Metric | 85.5, 88.5 | B MCU + Flash | |
| | **R52** | 10k | R_0402_1005Metric | 64.4, 85.0 | C 传感器 + GNSS | |
| | **R53** | 10k | R_0402_1005Metric | 64.3, 83.0 | C 传感器 + GNSS | |
| | **R54** | 10R | R_0402_1005Metric | 116.5, 77.0 | E 1090 接收链 | |
| | **R55** | 10k | R_0603_1608Metric | 57.4, 89.8 | C 传感器 + GNSS | |
| | **R56** | 10k | R_0603_1608Metric | 98.5, 90.0 | D 978 收发 | |
| | **R57** | 33R | R_0603_1608Metric | 104.0, 82.5 | E 1090 接收链 | |
| | **R58** | 10k | R_0603_1608Metric | 132.5, 108.2 | G 航空波段 AIRBAND | |
| | **R59** | 10k | R_0603_1608Metric | 132.5, 105.6 | G 航空波段 AIRBAND | |
| | **R60** | 0R | R_0603_1608Metric | 124.0, 101.5 | G 航空波段 AIRBAND | |

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|---|---|---|---|---|---|---|
| | R61 | 10k | R_0402_1005Metric | 131.1, 97.9 | G 航空波段 AIRBAND | |
| | L1 | 18nH 0402CS-18NXGRW | L_0402_1005Metric | 110.4, 59.8 | E 1090 接收链 | |
| | L2 | 33nH | L_0402_1005Metric | 57.7, 75.4 | C 传感器 + GNSS | |
| | L7 | 6.8uH | L_0805_2012Metric | 97.3, 94.7 | D 978 收发 | |
| | L8 | 100nH | L_0402_1005Metric | 119.3, 90.3 | D 978 收发 | |
| | L9 | 7.5nH | L_0402_1005Metric | 112.3, 94.8 | D 978 收发 | |
| | L10 | 27nH | L_0402_1005Metric | 110.9, 95.7 | D 978 收发 | |
| | L11 | 6.8nH | L_0402_1005Metric | 118.2, 97.5 | D 978 收发 | |
| | L12 | 6.8nH | L_0402_1005Metric | 118.2, 95.2 | D 978 收发 | |
| | L13 | 7.5nH | L_0402_1005Metric | 110.7, 92.2 | D 978 收发 | |
| | L14 | 100nH | L_0402_1005Metric | 101.1, 70.6 | E 1090 接收链 | |
| | L15 | 33nH | L_0402_1005Metric | 57.6, 68.7 | C 传感器 + GNSS | |
| | L16 | 1uH | L_Bourns-SRN4018 | 139.7, 83.6 | A 电源 | |
| | L17 | 4.7uH XEL4030-472MEC | L_Coilcraft_XxL4030 | 144.8, 95.0 | A 电源 | |
| | L18 | 100nH | L_0402_1005Metric | 126.0, 93.9 | G 航空波段 AIRBAND | |
| | D2 | TPESD8L3.3 0.3pFtyp 0.5pFmax | D_0402_1005Metric | 101.1, 72.6 | E 1090 接收链 | |
| | D3 | TPESD8L3.3 0.3pFtyp 0.5pFmax | D_0402_1005Metric | 117.5, 89.9 | D 978 收发 | |
| | D4 | TPESD8L3.3 0.3pFtyp 0.5pFmax | D_0402_1005Metric | 139.3, 75.4 | B MCU + Flash | |
| | D5 | TPESD8L3.3 0.3pFtyp 0.5pFmax | D_0402_1005Metric | 139.3, 77.8 | B MCU + Flash | |
| | D6 | TPESD8L3.3 0.3pFtyp 0.5pFmax | D_0402_1005Metric | 126.4, 89.3 | G 航空波段 AIRBAND | 有极性，认阴极标记带 |
| | F2 | 6V/200mA | Fuse_0805_2012Metric | 122.0, 91.0 | D 978 收发 | |
| | F3 | 6V/200mA | Fuse_0805_2012Metric | 98.3, 71.7 | E 1090 接收链 | |
| | F4 | 6V/200mA | Fuse_0805_2012Metric | 60.3, 75.9 | C 传感器 + GNSS | |
| | F5 | 6V/200mA | Fuse_0805_2012Metric | 60.3, 69.2 | C 传感器 + GNSS | |
| | Q2 | AO3401A | SOT-23 | 122.0, 93.9 | D 978 收发 | |
| | Q3 | AO3401A | SOT-23 | 95.3, 71.7 | E 1090 接收链 | |
| | Q4 | AO3401A | SOT-23 | 63.5, 75.9 | C 传感器 + GNSS | |
| | Q5 | AO3401A | SOT-23 | 63.5, 69.2 | C 传感器 + GNSS | |
| | U1 | ME6211C33M5 | SOT-23-5 | 74.7, 97.8 | A 电源 | 认 pin1（缺角/圆点） |
| | U2 | TPS7A2033PDBVR | SOT-23-5 | 82.3, 97.8 | A 电源 | 认 pin1（缺角/圆点） |
| | U3 | TPS7A2033PDBVR | SOT-23-5 | 89.2, 97.8 | A 电源 | 认 pin1（缺角/圆点） |
| | U4 | BNO085 | BNO085_LGA-28 | 59.7, 84.5 | C 传感器 + GNSS | 认 pin1（缺角/圆点） |
| | U5 | BMP388 | BMP388_LGA-10 | 61.0, 88.9 | C 传感器 + GNSS | 认 pin1（缺角/圆点） |
| | U6 | QMC5883P | LGA-16_3x3mm_P0.5mm | 59.7, 96.5 | C 传感器 + GNSS | 认 pin1（缺角/圆点） |
| | U7 | ATGM336H-6N-74 | ATGM336H_LCC-18 | 73.7, 89.1 | C 传感器 + GNSS | 认 pin1（缺角/圆点） |
| | U8 | RP2040 | QFN-56-1EP_7x7mm_P0.4mm_EP3. | 86.4, 76.9 | B MCU + Flash | 认 pin1（缺角/圆点） |
| | U9 | W25Q128JVSIQ | SOIC-8_5.3x5.3mm_P1.27mm | 93.7, 85.9 | B MCU + Flash | 认 pin1（缺角/圆点） |
| | U10 | CC1312R1F3RGZR | QFN-48-1EP_7x7mm_P0.5mm_EP5. | 104.5, 92.0 | D 978 收发 | 认 pin1（缺角/圆点） |
| | U11 | QPL9547TR7 | DFN-8-1EP_2x2mm_P0.5mm_EP0. | 107.7, 61.9 | E 1090 接收链 | 认 pin1（缺角/圆点） |
| | U12 | BGA2817 | SOT-363_SC-70-6 | 117.1, 62.5 | E 1090 接收链 | 认 pin1（缺角/圆点） |
| | U14 | AD8313ARMZ | MSOP-8_3x3mm_P0.65mm | 111.1, 73.4 | E 1090 接收链 | 认 pin1（缺角/圆点） |
| | U15 | TLV3501(TOKMAS) | SOT-23-6 | 100.6, 79.5 | E 1090 接收链 | 认 pin1（缺角/圆点） |
| | U16 | XA17-G4K | SOT-363_SC-70-6 | 98.0, 62.9 | E 1090 接收链 | 认 pin1（缺角/圆点） |
| | U17 | XA17-G4K | SOT-363_SC-70-6 | 68.6, 78.1 | C 传感器 + GNSS | 认 pin1（缺角/圆点） |
| | U18 | CH224K | CH224K_ESSOP-10 | 132.5, 68.6 | A 电源 | 认 pin1（缺角/圆点） |
| | U19 | SY6970 | QFN-24-1EP_4x4mm_P0.5mm_EP2. | 134.5, 83.7 | A 电源 | 认 pin1（缺角/圆点） |
| | U20 | SY7069 | TSOT-23-6 | 139.2, 95.0 | A 电源 | 认 pin1（缺角/圆点） |

| 贴 | 位号 | 值 / 型号 | 封装 | 位置 (x, y) | 阶段 | 备注 |
|---|---|---|---|---|---|---|
| | **U21** | BK4819QN32SC | QFN-32-1EP_4x4mm_P0.4mm_EP2. | 127.5, 97.6 | G 航空波段 AIRBAND | 认 pin1（缺角/圆点） |
| | **U22** | PCM1808PWR | TSSOP-14_4.4x5mm_P0.65mm | 122.8, 105.3 | G 航空波段 AIRBAND | 认 pin1（缺角/圆点） |
| | **FL1** | TA0970A | TA0970A_SMD3838-6 | 112.0, 65.5 | E 1090 接收链 | SAW 滤波器分输入输出，贴反不工作 |
| | **FL2** | TA0970A | TA0970A_SMD3838-6 | 118.4, 66.5 | E 1090 接收链 | SAW 滤波器分输入输出，贴反不工作 |
| | **Y1** | 12MHz CL=10pF ABM8-272-T3 | Crystal_SMD_3225-4Pin_3.2x2.5mm | 85.1, 69.1 | B MCU + Flash | 无源晶体，转 180° 等价，长边对长边即可 |
| | **Y2** | 48MHz ABM8W-7pF | Crystal_SMD_3225-4Pin_3.2x2.5mm | 108.3, 98.3 | D 978 收发 | 无源晶体，转 180° 等价，长边对长边即可 |
| | **Y3** | 32.768kHz FC-135 | Crystal_SMD_3215-2Pin_3.2x1.5mm | 111.6, 87.0 | D 978 收发 | |
| | **Y4** | 26MHz TCXO <=2.5ppm | Crystal_SMD_2520-4Pin_2.5x2.0mm | 135.1, 102.5 | G 航空波段 AIRBAND | 有源 TCXO：pin2=GND / pin4=Vcc，转 180° 直接短路、上电即烧。顶面无标记，倒角在底面 #1 角（手册 C0.2）——贴前翻过来认，或先用一颗定出「底面倒角↔顶面打标」的对应关系再批量贴；板上丝印的实心圆点指向 pin1 |
| | **J1** | J3_HAT_2x20_SMD排针 | PinHeader_2x20_P2.54mm_ | 92.6, 106.3 | F 对外接口 | |
| | **J2** | U.FL_GNSS_EXT(经尾线转SMA) | U.FL_Hirose_U.FL-R-SMT-1_Vertical | 54.0, 75.9 | C 传感器 + GNSS | |
| | **J4** | USB-C_16P | USB_C_Receptacle_HRO_TC-31-M-12 | 145.8, 73.0 | B MCU + Flash | |
| | **J5** | U.FL_978 | U.FL_Hirose_U.FL-R-SMT-1_Vertical | 118.3, 82.9 | D 978 收发 | |
| | **J6** | U.FL_1090_EXT(经尾线转SMA) | U.FL_Hirose_U.FL-R-SMT-1_Vertical | 106.5, 67.3 | E 1090 接收链 | |
| | **J7** | U.FL_IFA_TEST(π后调试口) | U.FL_Hirose_U.FL-R-SMT-1_Vertical | 90.0, 61.3 | E 1090 接收链 | |
| | **J8** | U.FL→内置patch | U.FL_Hirose_U.FL-R-SMT-1_Vertical | 54.0, 69.2 | C 传感器 + GNSS | |
| | **J9** | MX1.25WT-2P BAT | MX1.25WT-2P_1x02-1MP_P1.25mm_Horizontal | 144.8, 99.0 | A 电源 | |
| | **J10** | U.FL_VHF(经尾线转SMA) | U.FL_Hirose_U.FL-R-SMT-1_Vertical | 124.4, 82.9 | G 航空波段 AIRBAND | |
| | **ANT1** | IFA_1090 | ANT_IFA_1090MHz | 79.2, 57.8 | 不贴（板载天线） | 50.0mm外包络/48.5mm中心线；V4.0装盒实测 1082.5MHz、SWR 1.09 |
| | **ZP1** | DNP 并-天线侧 | C_0603_1608Metric | 79.2, 63.6 | 不贴（DNP） | 默认DNP；V4.0实测直通状态 |
| | **ZP2** | DNP 并-电台侧 | C_0603_1608Metric | 84.2, 63.6 | 不贴（DNP） | 默认DNP；历史HFSS 3.3pF仅作异常复调备选 |
| | **ZS1** | 0R 串 | L_0603_1608Metric | 81.7, 62.9 | E 1090 接收链 | 默认0R直通；V4.0实测即为此状态 |
| | **SW1** | RESET短接焊盘 | SolderJumper-2_P1.3mm_Open_Pad1.0x1. | 51.9, 77.5 | 不贴（短接焊盘） | |
| | **SW2** | BOOTSEL短接焊盘 | SolderJumper-2_P1.3mm_Open_Pad1.0x1. | 51.9, 81.0 | 不贴（短接焊盘） | |
| | **SW3** | TS24CA | SW_TS24CA_2x4_SideAct | 147.9, 89.1 | A 电源 | |
| | **TP1** | TestPoint | TestPoint_Pad_1.0x1.0mm | 51.9, 86.0 | 不贴（测试点焊盘） | |
| | **TP2** | TestPoint | TestPoint_Pad_1.0x1.0mm | 51.9, 88.1 | 不贴（测试点焊盘） | |
| | **TP3** | TestPoint | TestPoint_Pad_1.0x1.0mm | 51.9, 90.1 | 不贴（测试点焊盘） | |
| | **TP4** | TestPoint | TestPoint_Pad_1.0x1.0mm | 51.9, 92.1 | 不贴（测试点焊盘） | |
| | **TP5** | TestPoint | TestPoint_Pad_1.0x1.0mm | 51.9, 94.2 | 不贴（测试点焊盘） | |
| | **TP6** | TestPoint | TestPoint_Pad_1.0x1.0mm | 51.9, 96.2 | 不贴（测试点焊盘） | |
| | **TP7** | TestPoint | TestPoint_Pad_1.0x1.0mm | 51.9, 98.2 | 不贴（测试点焊盘） | |
| | **H1** | — | MountingHole_2.7mm_M2.5 | 54.0, 56.0 | 不贴（安装孔） | |
| | **H2** | — | MountingHole_2.7mm_M2.5 | 146.0, 56.0 | 不贴（安装孔） | |
| | **H3** | — | MountingHole_2.7mm_M2.5 | 54.0, 106.0 | 不贴（安装孔） | |
| | **H4** | — | MountingHole_2.7mm_M2.5 | 146.0, 106.0 | 不贴（安装孔） | |
| | **RT1** | NCP18XH103F03RB 10k NTC | R_0603_1608Metric | 54.8, 100.9 | A 电源 | |

# V4.7 BOM Verification Checklist (reference-designator order)

229 designators in total, 207 of them to be placed. Tick the box on the left after placing each part. Coordinate origin: PCB top-left corner, in mm. Chinese notes with fuller engineering context live in CHECKLIST-zh_CN.md.

Polarized / directional — reversed placement will not work | Not placed

| ✓ | Designator | Value / Model | Footprint | Position (x, y) | Stage | Notes |
|---|---|---|---|---|---|---|
| | **C1** | 10uF | C_0805_2012Metric | (70.8, 96.5) | A Power | |
| | **C2** | 1uF | C_0603_1608Metric | (85.9, 99.1) | A Power | |
| | **C3** | 1uF | C_0603_1608Metric | (70.8, 98.9) | A Power | |
| | **C4** | 1uF | C_0603_1608Metric | (78.7, 97.8) | A Power | |
| | **C5** | 1uF | C_0402_1005Metric | (85.9, 96.7) | A Power | |
| | **C6** | 1uF | C_0603_1608Metric | (92.1, 97.9) | A Power | |
| | **C10** | 100nF | C_0603_1608Metric | (61.0, 92.6) | C Sensors + GNSS | |
| | **C11** | 100nF | C_0603_1608Metric | (64.6, 87.2) | C Sensors + GNSS | |
| | **C12** | 100nF | C_0603_1608Metric | (56.5, 87.7) | C Sensors + GNSS | |
| | **C13** | 100nF | C_0603_1608Metric | (59.1, 92.6) | C Sensors + GNSS | |
| | **C14** | 100nF | C_0603_1608Metric | (64.6, 89.9) | C Sensors + GNSS | |
| | **C15** | 4.7uF | C_0603_1608Metric | (59.9, 99.3) | C Sensors + GNSS | |
| | **C16** | 100nF | C_0603_1608Metric | (64.6, 92.5) | C Sensors + GNSS | |
| | **C17** | 100nF | C_0603_1608Metric | (71.4, 81.0) | C Sensors + GNSS | |
| | **C18** | 10uF | C_0805_2012Metric | (74.4, 80.8) | C Sensors + GNSS | |
| | **C19** | 15pF C0G | C_0603_1608Metric | (88.4, 69.1) | B MCU + Flash | |
| | **C20** | 15pF C0G | C_0603_1608Metric | (79.8, 69.1) | B MCU + Flash | |
| | **C21** | 100pF | C_0402_1005Metric | (103.6, 59.7) | E 1090 Receive Chain | |
| | **C22** | 100nF | C_0603_1608Metric | (93.2, 75.8) | B MCU + Flash | |
| | **C23** | 100nF | C_0603_1608Metric | (91.8, 81.5) | B MCU + Flash | |
| | **C24** | 100nF | C_0603_1608Metric | (80.9, 80.2) | B MCU + Flash | |
| | **C25** | 100nF | C_0603_1608Metric | (83.2, 71.6) | B MCU + Flash | |
| | **C26** | 100nF | C_0603_1608Metric | (78.9, 74.6) | B MCU + Flash | |
| | **C27** | 100nF | C_0603_1608Metric | (77.9, 82.5) | B MCU + Flash | |
| | **C28** | 1uF | C_0603_1608Metric | (83.1, 83.3) | B MCU + Flash | |
| | **C29** | 1uF | C_0603_1608Metric | (80.9, 83.3) | B MCU + Flash | |
| | **C30** | 100pF | C_0402_1005Metric | (102.8, 67.5) | E 1090 Receive Chain | |
| | **C31** | 100pF C0G | C_0402_1005Metric | (111.3, 62.2) | E 1090 Receive Chain | |
| | **C32** | 100pF C0G | C_0402_1005Metric | (114.3, 62.2) | E 1090 Receive Chain | |
| | **C33** | 100pF C0G | C_0402_1005Metric | (120.4, 63.2) | E 1090 Receive Chain | |
| | **C34** | 100pF C0G | C_0402_1005Metric | (117.1, 73.4) | E 1090 Receive Chain | |
| | **C35** | 100pF C0G | C_0402_1005Metric | (115.1, 71.1) | E 1090 Receive Chain | |
| | **C36** | 100pF | C_0402_1005Metric | (106.0, 64.3) | E 1090 Receive Chain | |
| | **C37** | 100nF | C_0402_1005Metric | (113.9, 77.0) | E 1090 Receive Chain | |
| | **C38** | 100pF | C_0402_1005Metric | (111.6, 77.0) | E 1090 Receive Chain | |
| | **C39** | 100pF | C_0402_1005Metric | (118.2, 92.8) | D 978 Transceiver | |
| | **C40** | 3.6pF | C_0402_1005Metric | (110.7, 94.2) | D 978 Transceiver | |
| | **C41** | 2.7pF | C_0402_1005Metric | (115.9, 97.5) | D 978 Transceiver | |
| | **C42** | 6.2pF | C_0402_1005Metric | (115.9, 95.2) | D 978 Transceiver | |
| | **C43** | 3pF | C_0402_1005Metric | (115.9, 92.8) | D 978 Transceiver | |
| | **C44** | 3.6pF | C_0402_1005Metric | (112.8, 96.5) | D 978 Transceiver | |
| | **C45** | 100pF | C_0402_1005Metric | (110.7, 90.2) | D 978 Transceiver | |
| | **C46** | 3pF | C_0603_1608Metric | (104.9, 78.5) | E 1090 Receive Chain | |
| | **C47** | 200pF | C_0603_1608Metric | (104.9, 80.4) | E 1090 Receive Chain | |
| | **C48** | 1uF | C_0402_1005Metric | (107.4, 64.3) | E 1090 Receive Chain | |
| | **C49** | 100nF | C_0603_1608Metric | (76.7, 75.4) | E 1090 Receive Chain | |

| ✓ | Designator | Value / Model | Footprint | Position (x, y) | Stage | Notes |
|---|---|---|---|---|---|---|
| | **C51** | 1nF | C_0603_1608Metric | (77.0, 81.0) | E 1090 Receive Chain | |
| | **C52** | 100nF | C_0402_1005Metric | (99.6, 82.3) | E 1090 Receive Chain | |
| | **C53** | 100pF | C_0402_1005Metric | (93.6, 62.9) | E 1090 Receive Chain | See the Chinese companion for the detailed engineering note |
| | **C54** | 100pF | C_0402_1005Metric | (103.9, 61.7) | E 1090 Receive Chain | |
| | **C55** | 100pF | C_0603_1608Metric | (97.6, 66.5) | E 1090 Receive Chain | |
| | **C56** | 100pF | C_0603_1608Metric | (95.0, 66.5) | E 1090 Receive Chain | |
| | **C57** | 100pF | C_0402_1005Metric | (68.3, 81.0) | C Sensors + GNSS | |
| | **C58** | 100pF | C_0402_1005Metric | (57.7, 76.5) | C Sensors + GNSS | |
| | **C59** | 100pF | C_0402_1005Metric | (57.6, 69.8) | C Sensors + GNSS | |
| | **C60** | 22uF | C_0805_2012Metric | (99.3, 98.0) | D 978 Transceiver | |
| | **C61** | 100nF | C_0603_1608Metric | (102.7, 99.1) | D 978 Transceiver | |
| | **C62** | 100nF | C_0603_1608Metric | (105.4, 98.3) | D 978 Transceiver | |
| | **C63** | 100nF | C_0603_1608Metric | (107.1, 84.6) | D 978 Transceiver | |
| | **C64** | 100nF | C_0603_1608Metric | (100.2, 85.8) | D 978 Transceiver | |
| | **C65** | 100nF | C_0603_1608Metric | (99.0, 94.7) | D 978 Transceiver | |
| | **C66** | 22uF | C_0805_2012Metric | (102.9, 97.2) | D 978 Transceiver | |
| | **C67** | 1uF | C_0603_1608Metric | (103.6, 85.8) | D 978 Transceiver | |
| | **C68** | 18pF | C_0603_1608Metric | (113.2, 85.0) | D 978 Transceiver | |
| | **C69** | 18pF | C_0603_1608Metric | (110.1, 85.0) | D 978 Transceiver | |
| | **C70** | 47nF | C_0603_1608Metric | (135.7, 80.2) | A Power | |
| | **C71** | 4.7uF(带电源版) | C_0603_1608Metric | (133.4, 79.0) | A Power | |
| | **C72** | 10uF | C_1206_3216Metric | (130.9, 79.2) | A Power | |
| | **C73** | 10uF | C_0805_2012Metric | (141.3, 88.1) | A Power | |
| | **C74** | 10uF | C_0805_2012Metric | (143.3, 84.9) | A Power | |
| | **C75** | 10uF | C_0805_2012Metric | (139.1, 88.1) | A Power | |
| | **C76** | 22uF | C_1206_3216Metric | (135.2, 96.8) | A Power | |
| | **C77** | 22uF 16V X5R | C_1206_3216Metric | (141.2, 99.7) | A Power | |
| | **C78** | 1uF | C_0603_1608Metric | (134.3, 93.8) | A Power | |
| | **C79** | 1uF | C_0603_1608Metric | (130.6, 65.0) | A Power | |
| | **C80** | 1uF 25V | C_0603_1608Metric | (136.3, 74.6) | A Power | |
| | **C81** | 470pF C0G | C_0402_1005Metric | (115.2, 64.5) | E 1090 Receive Chain | |
| | **C82** | 100nF | C_0603_1608Metric | (91.4, 67.5) | B MCU + Flash | |
| | **C83** | 100nF | C_0603_1608Metric | (91.4, 69.8) | B MCU + Flash | |
| | **C84** | 1uF | C_0603_1608Metric | (83.3, 85.6) | B MCU + Flash | |
| | **C85** | 100nF | C_0603_1608Metric | (113.7, 80.5) | B MCU + Flash | |
| | **C86** | 100nF | C_0603_1608Metric | (121.4, 88.1) | B MCU + Flash | |
| | **C87** | 10nF C0G | C_0603_1608Metric | (83.5, 88.5) | B MCU + Flash | |
| | **C88** | 100pF | C_0603_1608Metric | (126.4, 90.7) | G Airband | |
| | **C89** | 18pF | C_0603_1608Metric | (128.7, 93.9) | G Airband | |
| | **C91** | 10uF | C_0805_2012Metric | (129.0, 108.2) | G Airband | |
| | **C92** | 10uF | C_0805_2012Metric | (121.3, 100.8) | G Airband | |
| | **C93** | 10uF | C_0805_2012Metric | (124.2, 109.4) | G Airband | |
| | **C94** | 100nF | C_0603_1608Metric | (120.5, 109.4) | G Airband | |
| | **C95** | 100nF | C_0603_1608Metric | (119.4, 100.1) | G Airband | |
| | **C96** | 100nF | C_0603_1608Metric | (128.0, 102.1) | G Airband | |
| | **C97** | 1uF | C_0603_1608Metric | (129.8, 102.1) | G Airband | |
| | **C98** | 100nF | C_0603_1608Metric | (131.6, 102.1) | G Airband | |
| | **C99** | 100nF | C_0402_1005Metric | (133.6, 100.1) | G Airband | |
| | **R1** | 10k | R_0603_1608Metric | (55.2, 84.5) | C Sensors + GNSS | |

| ✓ | Designator | Value / Model | Footprint | Position (x, y) | Stage | Notes |
|---|---|---|---|---|---|---|
| | **R2** | 4.7k | R_0603_1608Metric | (64.1, 97.8) | C Sensors + GNSS | |
| | **R3** | 4.7k | R_0603_1608Metric | (64.1, 95.2) | C Sensors + GNSS | |
| | **R4** | 1k | R_0603_1608Metric | (82.0, 69.1) | B MCU + Flash | |
| | **R5** | 10k | R_0603_1608Metric | (80.0, 73.2) | B MCU + Flash | |
| | **R6** | 10k | R_0603_1608Metric | (87.9, 86.4) | B MCU + Flash | |
| | **R7** | 5.1k DNP(带电源版) | R_0603_1608Metric | (141.4, 62.5) | Not placed (DNP) | |
| | **R8** | 5.1k DNP(带电源版) | R_0603_1608Metric | (139.4, 62.5) | Not placed (DNP) | |
| | **R9** | 27R | R_0603_1608Metric | (87.9, 83.3) | B MCU + Flash | |
| | **R10** | 27R | R_0603_1608Metric | (85.3, 83.3) | B MCU + Flash | |
| | **R11** | 3.32k | R_0402_1005Metric | (105.1, 59.7) | E 1090 Receive Chain | |
| | **R17** | 10k | R_0603_1608Metric | (122.0, 97.4) | D 978 Transceiver | |
| | **R18** | 10k | R_0603_1608Metric | (91.4, 71.4) | E 1090 Receive Chain | |
| | **R19** | 52.3R | R_0402_1005Metric | (115.1, 73.4) | E 1090 Receive Chain | |
| | **R21** | 0R | R_0603_1608Metric | (105.0, 72.6) | E 1090 Receive Chain | |
| | **R22** | 0R | R_0603_1608Metric | (100.3, 66.5) | E 1090 Receive Chain | |
| | **R23** | 0R | R_0603_1608Metric | (93.2, 77.2) | E 1090 Receive Chain | |
| | **R24** | 0R DNP(旁路外接) | R_0402_1005Metric | (101.1, 63.1) | Not placed (DNP) | |
| | **R25** | 0R DNP(旁路板载) | R_0402_1005Metric | (95.5, 62.9) | Not placed (DNP) | |
| | **R26** | 10k | R_0603_1608Metric | (67.3, 73.7) | C Sensors + GNSS | |
| | **R27** | 10k | R_0603_1608Metric | (67.3, 71.1) | C Sensors + GNSS | |
| | **R30** | 1k DNP | R_0402_1005Metric | (96.8, 77.2) | Not placed (DNP) | |
| | **R31** | 1k | R_0603_1608Metric | (101.6, 75.7) | E 1090 Receive Chain | |
| | **R32** | 10k | R_0603_1608Metric | (104.9, 75.7) | E 1090 Receive Chain | |
| | **R33** | 100k | R_0603_1608Metric | (100.9, 82.7) | E 1090 Receive Chain | |
| | **R34** | 10k | R_0603_1608Metric | (77.0, 73.2) | E 1090 Receive Chain | |
| | **R35** | 10k | R_0603_1608Metric | (77.0, 78.4) | E 1090 Receive Chain | |
| | **R36** | 1k DNP | R_0603_1608Metric | (96.5, 81.8) | Not placed (DNP) | |
| | **R37** | 0R | R_0603_1608Metric | (137.0, 66.4) | A Power | |
| | **R38** | 180R | R_0603_1608Metric | (133.1, 88.2) | A Power | |
| | **R39** | 5.62k | R_0603_1608Metric | (136.2, 88.2) | A Power | |
| | **R40** | 31.6k | R_0603_1608Metric | (134.6, 88.2) | A Power | |
| | **R41** | 470k | R_0603_1608Metric | (139.1, 98.6) | A Power | |
| | **R42** | 150k | R_0603_1608Metric | (137.4, 98.6) | A Power | |
| | **R43** | 1k | R_0603_1608Metric | (127.5, 65.0) | A Power | |
| | **R44** | 10k | R_0603_1608Metric | (137.0, 69.5) | A Power | |
| | **R45** | 10k | R_0603_1608Metric | (129.2, 84.0) | A Power | |
| | **R46** | 10k | R_0603_1608Metric | (129.2, 86.5) | A Power | |
| | **R47** | 10k | R_0603_1608Metric | (98.5, 91.7) | D 978 Transceiver | |
| | **R48** | 0R | R_0603_1608Metric | (126.8, 67.0) | A Power | |
| | **R49** | 0R | R_0603_1608Metric | (126.8, 69.4) | A Power | |
| | **R50** | 30k | R_0603_1608Metric | (86.5, 92.8) | B MCU + Flash | |
| | **R51** | 10k | R_0603_1608Metric | (85.5, 88.5) | B MCU + Flash | |
| | **R52** | 10k | R_0402_1005Metric | (64.4, 85.0) | C Sensors + GNSS | |
| | **R53** | 10k | R_0402_1005Metric | (64.3, 83.0) | C Sensors + GNSS | |
| | **R54** | 10R | R_0402_1005Metric | (116.5, 77.0) | E 1090 Receive Chain | |
| | **R55** | 10k | R_0603_1608Metric | (57.4, 89.8) | C Sensors + GNSS | |
| | **R56** | 10k | R_0603_1608Metric | (98.5, 90.0) | D 978 Transceiver | |
| | **R57** | 33R | R_0603_1608Metric | (104.0, 82.5) | E 1090 Receive Chain | |
| | **R58** | 10k | R_0603_1608Metric | (132.5, 108.2) | G Airband | |
| | **R59** | 10k | R_0603_1608Metric | (132.5, 105.6) | G Airband | |

| ✓ | Designator | Value / Model | Footprint | Position (x, y) | Stage | Notes |
|---|---|---|---|---|---|---|
| | **R60** | 0R | R_0603_1608Metric | (124.0, 101.5) | G Airband | |
| | **R61** | 10k | R_0402_1005Metric | (131.1, 97.9) | G Airband | |
| | **L1** | 18nH 0402CS-18NXGRW | L_0402_1005Metric | (110.4, 59.8) | E 1090 Receive Chain | |
| | **L2** | 33nH | L_0402_1005Metric | (57.7, 75.4) | C Sensors + GNSS | |
| | **L7** | 6.8uH | L_0805_2012Metric | (97.3, 94.7) | D 978 Transceiver | |
| | **L8** | 100nH | L_0402_1005Metric | (119.3, 90.3) | D 978 Transceiver | |
| | **L9** | 7.5nH | L_0402_1005Metric | (112.3, 94.8) | D 978 Transceiver | |
| | **L10** | 27nH | L_0402_1005Metric | (110.9, 95.7) | D 978 Transceiver | |
| | **L11** | 6.8nH | L_0402_1005Metric | (118.2, 97.5) | D 978 Transceiver | |
| | **L12** | 6.8nH | L_0402_1005Metric | (118.2, 95.2) | D 978 Transceiver | |
| | **L13** | 7.5nH | L_0402_1005Metric | (110.7, 92.2) | D 978 Transceiver | |
| | **L14** | 100nH | L_0402_1005Metric | (101.1, 70.6) | E 1090 Receive Chain | |
| | **L15** | 33nH | L_0402_1005Metric | (57.6, 68.7) | C Sensors + GNSS | |
| | **L16** | 1uH | L_Bourns-SRN4018 | (139.7, 83.6) | A Power | |
| | **L17** | 4.7uH XEL4030-472MEC | L_Coilcraft_XxL4030 | (144.8, 95.0) | A Power | |
| | **L18** | 100nH | L_0402_1005Metric | (126.0, 93.9) | G Airband | |
| | **D2** | TPESD8L3.3 0.3pFtyp 0.5pFmax | D_0402_1005Metric | (101.1, 72.6) | E 1090 Receive Chain | Polarized; follow the cathode marking |
| | **D3** | TPESD8L3.3 0.3pFtyp 0.5pFmax | D_0402_1005Metric | (117.5, 89.9) | D 978 Transceiver | Polarized; follow the cathode marking |
| | **D4** | TPESD8L3.3 0.3pFtyp 0.5pFmax | D_0402_1005Metric | (139.3, 75.4) | B MCU + Flash | Bidirectional low-capacitance USB ESD protector; no polarity |
| | **D5** | TPESD8L3.3 0.3pFtyp 0.5pFmax | D_0402_1005Metric | (139.3, 77.8) | B MCU + Flash | Bidirectional low-capacitance USB ESD protector; no polarity |
| | **D6** | TPESD8L3.3 0.3pFtyp 0.5pFmax | D_0402_1005Metric | (126.4, 89.3) | G Airband | See the Chinese companion for the detailed engineering note |
| | **F2** | 6V/200mA | Fuse_0805_2012Metric | (122.0, 91.0) | D 978 Transceiver | |
| | **F3** | 6V/200mA | Fuse_0805_2012Metric | (98.3, 71.7) | E 1090 Receive Chain | |
| | **F4** | 6V/200mA | Fuse_0805_2012Metric | (60.3, 75.9) | C Sensors + GNSS | |
| | **F5** | 6V/200mA | Fuse_0805_2012Metric | (60.3, 69.2) | C Sensors + GNSS | |
| | **Q2** | AO3401A | SOT-23 | (122.0, 93.9) | D 978 Transceiver | |
| | **Q3** | AO3401A | SOT-23 | (95.3, 71.7) | E 1090 Receive Chain | |
| | **Q4** | AO3401A | SOT-23 | (63.5, 75.9) | C Sensors + GNSS | |
| | **Q5** | AO3401A | SOT-23 | (63.5, 69.2) | C Sensors + GNSS | |
| | **U1** | ME6211C33M5 | SOT-23-5 | (74.7, 97.8) | A Power | Observe pin 1 / signal direction |
| | **U2** | TPS7A2033PDBVR | SOT-23-5 | (82.3, 97.8) | A Power | Observe pin 1 / signal direction |
| | **U3** | TPS7A2033PDBVR | SOT-23-5 | (89.2, 97.8) | A Power | Observe pin 1 / signal direction |
| | **U4** | BNO085 | BNO085_LGA-28 | (59.7, 84.5) | C Sensors + GNSS | Observe pin 1 / signal direction |
| | **U5** | BMP388 | BMP388_LGA-10 | (61.0, 88.9) | C Sensors + GNSS | Observe pin 1 / signal direction |
| | **U6** | QMC5883P | LGA-16_3x3mm_P0.5mm | (59.7, 96.5) | C Sensors + GNSS | Observe pin 1 / signal direction |
| | **U7** | ATGM336H-6N-74 | ATGM336H_LCC-18 | (73.7, 89.1) | C Sensors + GNSS | Observe pin 1 / signal direction |
| | **U8** | RP2040 | QFN-56-1EP_7x7mm_P0.4m | (86.4, 76.9) | B MCU + Flash | Observe pin 1 / signal direction |
| | **U9** | W25Q128JVSIQ | SOIC-8_5.3x5.3mm_P1.27 | (93.7, 85.9) | B MCU + Flash | Observe pin 1 / signal direction |
| | **U10** | CC1312R1F3RGZR | QFN-48-1EP_7x7mm_P0.5m | (104.5, 92.0) | D 978 Transceiver | Observe pin 1 / signal direction |
| | **U11** | QPL9547TR7 | DFN-8-1EP_2x2mm_P0.5mm | (107.7, 61.9) | E 1090 Receive Chain | Observe pin 1 / signal direction |
| | **U12** | BGA2817 | SOT-363_SC-70-6 | (117.1, 62.5) | E 1090 Receive Chain | Observe pin 1 / signal direction |
| | **U14** | AD8313ARMZ | MSOP-8_3x3mm_P0.65mm | (111.1, 73.4) | E 1090 Receive Chain | Observe pin 1 / signal direction |
| | **U15** | TLV3501(TOKMAS) | SOT-23-6 | (100.6, 79.5) | E 1090 Receive Chain | Observe pin 1 / signal direction |
| | **U16** | XA17-G4K | SOT-363_SC-70-6 | (98.0, 62.9) | E 1090 Receive Chain | Observe pin 1 / signal direction |
| | **U17** | XA17-G4K | SOT-363_SC-70-6 | (68.6, 78.1) | C Sensors + GNSS | Observe pin 1 / signal direction |
| | **U18** | CH224K | CH224K_ESSOP-10 | (132.5, 68.6) | A Power | Observe pin 1 / signal direction |
| | **U19** | SY6970 | QFN-24-1EP_4x4mm_P0.5m | (134.5, 83.7) | A Power | Observe pin 1 / signal direction |

| ✓ | Designator | Value / Model | Footprint | Position (x, y) | Stage | Notes |
|---|---|---|---|---|---|---|
| | **U20** | SY7069 | TSOT-23-6 | (139.2, 95.0) | A Power | Observe pin 1 / signal direction |
| | **U21** | BK4819QN32SC | QFN-32-1EP_4x4mm_P0.4m | (127.5, 97.6) | G Airband | Observe pin 1 / signal direction |
| | **U22** | PCM1808PWR | TSSOP-14_4.4x5mm_P0.65 | (122.8, 105.3) | G Airband | Observe pin 1 / signal direction |
| | **FL1** | TA0970A | TA0970A_SMD3838-6 | (112.0, 65.5) | E 1090 Receive Chain | See the Chinese companion for the detailed engineering note |
| | **FL2** | TA0970A | TA0970A_SMD3838-6 | (118.4, 66.5) | E 1090 Receive Chain | See the Chinese companion for the detailed engineering note |
| | **Y1** | 12MHz CL=10pF ABM8-272-T3 | Crystal_SMD_3225-4Pin_ | (85.1, 69.1) | B MCU + Flash | Passive crystal; 180-degree rotation is electrically equivalent |
| | **Y2** | 48MHz ABM8W-7pF | Crystal_SMD_3225-4Pin_ | (108.3, 98.3) | D 978 Transceiver | Passive crystal; 180-degree rotation is electrically equivalent |
| | **Y3** | 32.768kHz FC-135 | Crystal_SMD_3215-2Pin_ | (111.6, 87.0) | D 978 Transceiver | |
| | **Y4** | 26MHz TCXO <=2.5ppm | Crystal_SMD_2520-4Pin_ | (135.1, 102.5) | G Airband | ACTIVE TCXO: pin2=GND / pin4=Vcc — rotating it 180 degrees shorts them and destroys the part on power-up. There is NO marking on the top face; the 0.2mm chamfer is on the BOTTOM at corner #1 (datasheet C0.2) — check it before placing, or use one unit to establish the bottom-chamfer / top-marking relationship first. The filled silkscreen dot on the board points at pin 1 |
| | **J1** | J3_HAT_2x20_SMD排针 | PinHeader_2x20_P2.54mm | (92.6, 106.3) | F External Interface | |
| | **J2** | U.FL_GNSS_EXT(经尾线转SMA) | U.FL_Hirose_U.FL-R-SMT | (54.0, 75.9) | C Sensors + GNSS | |
| | **J4** | USB-C_16P | USB_C_Receptacle_HRO_T | (145.8, 73.0) | B MCU + Flash | |
| | **J5** | U.FL_978 | U.FL_Hirose_U.FL-R-SMT | (118.3, 82.9) | D 978 Transceiver | |
| | **J6** | U.FL_1090_EXT(经尾线转SMA) | U.FL_Hirose_U.FL-R-SMT | (106.5, 67.3) | E 1090 Receive Chain | |
| | **J7** | U.FL_IFA_TEST(π后调试口) | U.FL_Hirose_U.FL-R-SMT | (90.0, 61.3) | E 1090 Receive Chain | |
| | **J8** | U.FL→内置patch | U.FL_Hirose_U.FL-R-SMT | (54.0, 69.2) | C Sensors + GNSS | |
| | **J9** | MX1.25WT-2P BAT | MX1.25WT-2P_1x02-1MP_P | (144.8, 99.0) | A Power | |
| | **J10** | U.FL_VHF(经尾线转SMA) | U.FL_Hirose_U.FL-R-SMT | (124.4, 82.9) | G Airband | |
| | **ANT1** | IFA_1090 | ANT_IFA_1090MHz | (79.2, 57.8) | Not placed (on-board antenna; per-cut measured 50.0mm outer envelope / 48.5mm centerline span) | Per-cut measured 50.0mm outer envelope / 48.5mm centerline span |
| | **ZP1** | DNP 并-天线侧 | C_0603_1608Metric | (79.2, 63.6) | Not placed (DNP) | DNP by default; see the Chinese companion for tuning details |
| | **ZP2** | DNP 并-电台侧 | C_0603_1608Metric | (84.2, 63.6) | Not placed (DNP) | DNP by default; see the Chinese companion for tuning details |
| | **ZS1** | 0R 串 | L_0603_1608Metric | (81.7, 62.9) | E 1090 Receive Chain | See the Chinese companion for the detailed engineering note |
| | **SW1** | RESET短接焊盘 | SolderJumper-2_P1.3mm_ | (51.9, 77.5) | Not placed (solder-bridge pad) | |
| | **SW2** | BOOTSEL短接焊盘 | SolderJumper-2_P1.3mm_ | (51.9, 81.0) | Not placed (solder-bridge pad) | |
| | **SW3** | TS24CA | SW_TS24CA_2x4_SideAct | (147.9, 89.1) | A Power | |
| | **TP1** | TestPoint | TestPoint_Pad_1.0x1.0m | (51.9, 86.0) | Not placed (test-point pad) | |
| | **TP2** | TestPoint | TestPoint_Pad_1.0x1.0m | (51.9, 88.1) | Not placed (test-point pad) | |
| | **TP3** | TestPoint | TestPoint_Pad_1.0x1.0m | (51.9, 90.1) | Not placed (test-point pad) | |
| | **TP4** | TestPoint | TestPoint_Pad_1.0x1.0m | (51.9, 92.1) | Not placed (test-point pad) | |
| | **TP5** | TestPoint | TestPoint_Pad_1.0x1.0m | (51.9, 94.2) | Not placed (test-point pad) | |
| | **TP6** | TestPoint | TestPoint_Pad_1.0x1.0m | (51.9, 96.2) | Not placed (test-point pad) | |
| | **TP7** | TestPoint | TestPoint_Pad_1.0x1.0m | (51.9, 98.2) | Not placed (test-point pad) | |
| | **H1** | — | MountingHole_2.7mm_M2. | (54.0, 56.0) | Not placed (mounting hole) | |
| | **H2** | — | MountingHole_2.7mm_M2. | (146.0, 56.0) | Not placed (mounting hole) | |
| | **H3** | — | MountingHole_2.7mm_M2. | (54.0, 106.0) | Not placed (mounting hole) | |
| | **H4** | — | MountingHole_2.7mm_M2. | (146.0, 106.0) | Not placed (mounting hole) | |

| ✓ | Designator | Value / Model | Footprint | Position (x, y) | Stage | Notes |
|---|---|---|---|---|---|---|
| | **RT1** | NCP18XH103F03RB<br>10k NTC | R_0603_1608Metric | (54.8, 100.9) | A Power | |